



Port Said University
Faculty of Science
Environmental Science Department



Impact of dairy wastewater on Nile Tilapia (*O. niloticus*) : Case Study Industrial zone Gamasa, El Dakahleya , Egypt.

By

Omar Khaled

Omar El Essawy

Aseel Saad

Nada Farag

Shaimaa Shitewy

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Faculty of science - Port Said University

Supervised by

Dr. Manal El Barbary

Dr. Ahmed Abdelaal

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Contents

Abstract.....	5
1. Introduction.....	8
1.1 Water Quality and Sustainability for Humanity's Survival.....	8
1.2 Water resources in Egypt.	9
1.3 Water Expenditure Per Capita.....	12
2- Water Pollution.....	15
2.1 Water pollution can occur from two sources.	16
2.2 - The Presence of Pollutants in the Aquatic Environment	17
2.3- Heavy Metals	18
2.4- Pesticides.....	19
3- Water pollution from Dairy industry	22
3.1 Dairy industry.....	22
3.2 Characteristics of dairy industry wastewaters.....	26
3.3 Effects of Effluents: Environmental Effects.	28
3.4 The impact of dairy industry wastewater on human health.	31
3.6 Health Impacts:	34
3.7 Water Quality Degradation	38
4- Evaluation the dairy wastewater on tilapia (<i>O. niloticus</i>).....	41
4.1 Material and methods :.....	41
Stress markers assessments	42
4.2 Results and Discussion.....	43
5- Conclusion	49
Recommendations	52
References	54

List of tables

Table 1 Water supplies and demands in Egypt (109 m ³ /yr) - * After the Gungli Canal Based on Abdel-Dayem (1994) and El-Kassas.(١٩٩٨)	10
Table 2 Degradation of the yearly water share per capita over a century.....	13
Table 3 Sources of pollution in dairy industry (EL-Tawabty et.al. 2022).....	25
Table 4 Typical Characteristics of dairy industry wastewaters reported by various authors are given in Table	27
Table 5 Table: Experimental fish groups.....	42
Table 6 The chemical composition of the DWW %is presented in Table.....	43
Table 7 : Mortality rate (%) of fish exposed to different dilutions of dairy wastes.....	43
Table 8 variations in the cortisol and glucose levels	44

Abstract

Globally, the dairy sector is a substantial industry amounting to 900 million tons of milk being produced annually. However, this large-scale production raises concerns about the resulting wastewater. It is estimated that the volume of wastewater generated during milk processing is 2.5 times greater than the volume of milk processed, amounting to a staggering 2.25 billion tonnes of dairy wastewater produced annually.

This study investigates the impact of dairy industry effluents on the aquatic environment, with a particular focus on Nile tilapia (*Oreochromis niloticus*). The increasing discharge of untreated or inadequately treated wastewater from dairy operations poses significant environmental challenges, particularly for aquatic ecosystems. A controlled experimental setup was designed to evaluate the effects of varying concentrations of dairy effluent on Nile tilapia. Key parameters such as water quality, fish behavior, and mortality rates were monitored.

The results demonstrated a direct correlation between the concentration of dairy effluent and the mortality rate of Nile tilapia, indicating that higher levels of drainage significantly increased fish mortality.

This study underscores the urgent need for effective wastewater management practices in the dairy industry to protect marine life and ensure sustainable aquatic ecosystems.

Chapter 1

Introduction

1. Introduction

1.1 Water Quality and Sustainability for Humanity's Survival.

Water can be considered as the most important natural resource that can be utilized by man to develop his prosperity as well as his essential needs. Water quality management, water pollution control and environmental protection are the main issues to save our future. The total quantity of water on the earth is about $13.2 \times 10^9 \text{ km}^3$, 97% of it is marine water, the freshwater is only 2.7%. Of the latter, 90% is either frozen in the North and South Poles or very deep groundwater at a depth of more than 700 m.

Thus, they are not available to man. The total amount of water in all the world rivers are about 48.000 km^3 , humans use less than $4.000 \text{ km}^3 / \text{yr}$, only. It was estimated that if we could possibly use 1% more out of our freshwater, we should be able to cultivate all the deserts and the arid and semi-arid land on the earth that cover 41% of the total land. Thus, we should not have any food deficiency on the earth. It is worth mentioning that the amount of groundwater in the world is estimated to be about 22 million km^3 .

In a survey of world freshwater, it has been reported that Egypt is among the ten countries to be scarce of water by the year 2025 due to the rapidly increasing population (Engelman and Le Roy, 1993), at present about 67.7 million inhabitants.

As a matter of fact, the full control of the river consistently safeguards Egypt against draught and heavy floods. For the time being, deteriorating water quality is a serious threat in countries with a scarcity of water resources. It does not only diminish the country's chance to sustainable development but it also threatens

public health with spreading infectious diseases. Water-related diseases are the most common causes of infants' mortality in the developing countries.

1.2 Water resources in Egypt.

1.2.1 Nile water: According to an agreement between Egypt and Sudan (1959) the Nile water budget is $18.5 \times 10^9 \text{ m}^3$ to Sudan and $55.5 \times 10^9 \text{ m}^3$ to Egypt (Dijkman, 1993). Nile water comprises about 97% of the renewable water supplies in Egypt. Table 1 demonstrates the water supplies and demands in Egypt. The total cultivated area in Egypt is less than 4%.

The sources of Egyptian Nile water supplies are in the Ethiopian (83%) and Equatorial Plateaus (17%). The yield of the former can be divided as follows: 13%, 58% and 12% from Subat, Blue Nile and Otbara River, respectively. In case of the latter, it is amazing that of its huge resources (about $110 \times 10^9 \text{ m}^3$ /yr) only $30 \times 10^9 \text{ m}^3$ /yr reach the Victoria Nile branch. The rest is mainly lost by evaporation. The total amount of water collected south of Sudan ($33 \times 10^9 \text{ m}^3$ /yr) spreads over the giant 700 km² swamps.

These swamps receive water from Bahr-El-Ghazal as well as from Bahr-El-Arab. The former is extended over more than 160 km to Lake Nu. The output to the White Nile is only $15 \times 10^9 \text{ m}^3$ /yr. That is why the construction of Gunglie Canal can ensure the supply to Egypt as well as North Sudan in an amount of a few milliards cubic meters of water originally lost by evaporation along the big distance. Hence, the cooperation between the Nile valley countries is essential to protect such a vital source of water (Hagras, 1988; Said, 1993).

I. Water supplies	1990	2000	2025
Nile water			
Groundwater:	55.5	57.5*	57.5*
In the Delta and New Valley			
In the desert	2.6	5.1	6.3
Reuse of agricultural drainage	0.5		
water	4.7	7.0	8.0
Treated sewage water	0.2	1.1	2.4
Management and saving wasted	—	1.0	—
water			
Total	63.5	71.7	74.2
II. Water demands			
Agriculture	49.7	59.9	61.5
Households	3.1	3.1	5.1
Industry	4.6	6.1	8.6
Navigation	1.8	0.3	0.4
Total	59.2	69.4	75.6

Table 1 Water supplies and demands in Egypt (109 m³/yr) - * After the Gungli Canal Based on Abdel-Dayem (1994) and El-Kassas. (١٩٩٨)

1.2.2 Groundwater: The major groundwater aquifers (Table 1) stretch in the western desert and Sinai, beneath and west of the Delta (Natroun Valley and Cairo-Alexandria desert road), Salheyia, beneath Upper Egypt (Engelman and Le Roy, 1993). This water is renewable because it originates from the leakage of Nile and drainage waters.

The percolation of excess irrigation water and extraction of groundwater in the rest sectors of the Delta largely govern the process (AbdelDayem, 1994). Seawater intrusion along the northern sector of the Delta aquifer renders it nearly saline. The top layer is fresh over the saline water. Therefore, overconsumption of such water affects its quality and salinity. On the other hand, fresh water within the northern frontiers of the Delta plays an important role in safeguarding the region against Mediterranean Sea saline water.

Rice cultivation within the northern sector plays an effective role in protecting fresh groundwater (Chapman, 1992).

The quality of groundwater is so far evaluated on an intermittent basis, predicting signs of pollution within a 30-meter depth (Abdel-Dayem, 1994).

(2) There are some geological formations within the Egyptian desert that carry groundwater. Among the most important geological layers is the Nobian limestone in the western desert that supplies the New Valley' s Oasis.

The water in a layer is in the depth of only 60– 100 m around the area of East-Oweinat. The Dakhla, Kharga and Siwa Oases are within similar geological formations, but the water is usually at variable depths. It is worth mentioning that these groundwaters are of non-renewable fossil origin. Thus, bearing in mind the rights of next generations, optimizing their use in an economic way is very important.

1.2.3 Rainwater

Within the western part from Alexandria city to El-Sallum, the northern coast receives a modest amount of winter rainfalls with a mean level of 150 mm/yr. It diminishes to 100 mm/yr to the east in El-Arish region then rises up again to 250 mm/yr eastern of El-Arish at Rafah, northeast Sinai.

The water available is merely sufficient for pastoral purposes, to some extent for seasonal agriculture (barley) especially in excessively rainy years, for cultivating some olive and fig in the western and peach in the eastern zones. Coastal sand dunes and valley sediments that receive rainfall accommodate shallow wells for drinking water. In Sinai, some reservoirs have been constructed to collect heavy rainfalls in the valleys.

1.3 Water Expenditure Per Capita.

Considering the drinking water resources, the individual's expenditure in Egypt was around 1000 m³ /yr at a population size of 58 million. By the year 2000 it decreased to 957 m³ /yr which can be divided as 7% and 93% for the domestic use and the industrial and agricultural uses, respectively. Compared to the minimum demand required per individual (1300 m³) it can be seen that Egypt is far below that level. It is worth mentioning that the per capita water income in the USA, India, China, and the international level are 10.000, 2.430, 2.520, and 2.500 m³ , respectively. Thus, it can be concluded that the Egyptian water expenditure is about 38% of the international level. Table 2 shows the degradation of the water share per capita over the period of a century.

To regulate the Nile flow to the Delta, the construction of El-Quanatir Barrages was started in 1843 and was completed in 1861. The purpose was to increase the agricultural area and the full use of all the available water. The system of continuous irrigation was achieved by the year 1890 throughout the Delta (El-Noberg, 1993).

Further ambitious progress was made by constructing the Aswan High Dam, which was completed in 1970 creating the largest artificial lake in the world. cshows the high dam storage capacity.

The advantages of this gigantic project can be summarized as follows:

- reclamation of 336.1×10^3 ha during the 1960' s, and 415.5×10^3 ha during the 1980' s, most of them in Upper Egypt;
- production of 1010 kW/yr of electric power (about 53% of the energy produced in 1977);

- protection of Egypt from high floods threatening the country in some years;
- regulation and prevention of unnecessary loss of water which is limited and determined by a treaty between Egypt and Sudan; and
- full control of Nile in Lower Egypt.

Year	Water quantity per capita (m³)	Crop area/capita (m²)	Cultivated land area per capita (m²)
1897	5084	2914	2226
1907	4414	2876	2017
1917	3854	2575	1738
1927	3484	2556	1642
1937	3103	2276	1401
1947	2604	2004	1275
1960	1893	1656	950
1970	1713	1373	759
1986	1138	1080	588
1992	981	941	535
2000	957	924	523

Table 2 Degradation of the yearly water share per capita over a century

Chapter 2

Water Pollution

2- Water Pollution

Water is one of the renewable resources essential for sustaining all forms of life, food production, economic development, and for general well being. It is impossible to substitute for most of its uses, difficult to de pollute, expensive to transport, and it is truly a unique gift to mankind from nature. Water is also one of the most manageable natural resources as it is capable of diversion, transport, storage, and recycling. All these properties impart to water its great utility for human beings. The surface water and groundwater resources of the country play a major role in agriculture, hydropower generation, livestock production, industrial activities, forestry, fisheries, navigation, recreational activities etc. The freshwater ecosystems of the world comprise only about 0.5% of the earth's surface and have a volume of $2.84 \times 10^5 \text{ Km}^3$. Rivers constitute an insignificant amount (0.1%) of the land surface. Only 0.01% of the waters of the earth occur in river channels. In spite of these low quantities, running waters are of enormous significance (Wetzel, 2001).

In the last few decades, there has been a tremendous increase in the demand for freshwater due to rapid growth of population and the accelerated pace of industrialization (Ramakrishnaiah et al., 2009). Human health is threatened by most of the agricultural development activities particularly in relation to excessive application of fertilizers and unsanitary conditions (Okeke and Igboanua, 2003). Anthropogenic activities related to extensive urbanization, agricultural practices, industrialization, and population expansion have led to water quality deterioration in many parts of the world (Baig et al. 2009, Mian et al., 2010, Wang et al., 2010).

In addition, deficient water resources have increasingly restrained water pollution control and water quality improvement (Bu et al., 2010). Water pollution has been a research focus for government and scientists. Therefore, protecting river

water quality is extremely urgent because of serious water pollution and global scarcity water resources.

2.1 Water pollution can occur from two sources.

1. Point source

Point sources of pollution are those which have direct identifiable source. Example includes pipe attached to a factory, oil spill from a tanker, effluents coming out from industries. Point sources of pollution include wastewater effluent (both municipal and industrial) and storm sewer discharge and affect mostly the area near it.

2. non-point source :

non-point sources of pollution are those which arrive from different sources of origin and number of ways by which contaminants enter groundwater or surface water and arrive in the environment from different non identifiable sources. Examples are runoff from agricultural fields, urban waste etc.

Sometimes pollution that enters the environment in one place has an effect hundreds or even thousands of miles away. This is known as transboundary pollution. One example is the radioactive waste that travels through the oceans from nuclear reprocessing plants to nearby countries.

Water pollutants may be i) Organic and ii) Inorganic water pollutant .

1 .Organic water pollutants: They comprise of insecticides and herbicides, organohalides and other forms of chemicals; bacteria from sewage and livestock farming; food processing wastes; pathogens; volatile organic compounds etc.

2. Inorganic water pollutants: They may arise from heavy metals from acid mine drainage; silt from surface run-off, logging, slash and burning practices and land filling; fertilizers from agricultural run-off which include nitrates and phosphates etc. and chemical waste from industrial effluents.

2.2 - The Presence of Pollutants in the Aquatic Environment

It is well confirmed that pollutants enter the aquatic environment at different times. Pollutants include the following:

- 1- **Hydrocarbons:** e.g. polycyclic aromatic hydrocarbons (PAHs) Nwaichi & Ntorgbo (2016)
- 2- **Pesticides:** e.g. *Organophosphorus pesticides* (Roberts and Reigart, 2013) and *Organochlorinated pesticides* (Misumi et al., 2005).
- 3- **Heavy Metals:** e.g. arsenic, copper, zinc, cadmium, lead, and mercury (Okocha and Adedeji 2011) which are discharged in industrial effluents.
- 4- **Organic sewage,** as faecal debris that may contain large number of bacteria (Islam & Tanaka 2004).
- 5- **Inorganic nitrogen:** as ammonia, that are produced from the activities of fish farming and agricultural runoff .
- 6- **Plastics** (Barriáa 2020)
- 7- **Toxins,** such as from the breakdown of dinoflagellate blooms (Noga et al. 1996).
- 8- **Industrial pollution**

Farmed fish live in intimate contact with their environment and continuously exposed to several environmental stressors such as handling, crowding, and pathogens beside the pollution leading to numerous physiological alterations that finally caused depression of immunity and increase outbreaks of infection (Low and Sin 1998).

Fish are exposed to pesticides and their residues and heavy metals by various routes such as run-offs or spray drifting from agricultural fields, fish is an excellent guard to assay the toxicity of pollutants. Pesticides degrade the quality of water bodies followed by affecting the permanent organisms.

2.3- Heavy Metals

Heavy metals are dangerous pollutants, which intimidate aquatic ecosystems during their oxidative stress. Environmental pollution is caused by the mining and liquid wastes of metal industry effluents. They have high levels of toxic effects, they accumulated in fish causing serious effects on fish physiology and metabolism due to they are not degradable or eliminated (Scotta and Sloman 2004).

Some of these metals have found application in several areas of human life, such as agriculture, pharmacy, cosmetics, industry, medicine, etc. Contamination of water with these pollutants may be led to harmful consequences on the living organisms including fish of these environments (Farombi et al., (2007) & Vutukuru 2005)

The results of many metal poisoning depend not only on the physical chemical factors of water, but also on the quantity and duration of the action. Fish are highly sensitive to pollution and are considered as biomarkers for water quality (Carnston et al., 1996)

Heavy metals pose a threat to aquatic ecosystems caused by their bioaccumulation. Pollutants can be accumulated and biomagnified in water, sediment and the aquatic food chain (Nadmitov et al., 2015, Sloman 2007).

Fish is a good source of protein for humans, their poisoning with heavy metals can also risk human health (Yi et al 2011, Özparlak et al 2012).

The pollution of aquatic environment is one of the main risks affecting aquatic life in world. One of these threats is heavy metal pollution (Ahmed et al. 2013), where the exposure of heavy metals causes several deleterious health risk such as immunotoxicity, genotoxicity, and nephrotoxicity in living animals (Cobbina et al. 2015)

2.3.2. Influence of pollutants upon physiological biochemical parameters of fish.

- . *Effects of heavy metals on fish*

The biological cycles of heavy metals are very slow. The most dangerous elements in the ecotoxicological aspect are Hg, Cd, Cu, and Zn (Spry and Wiener 1991). Some micronutrients, e.g. Cu and Zn, have biological functions, in which they included in the active centers of enzymes and act as regulators of many biochemical processes. Meanwhile, the biological functions of Cd and Hg are not well known. However, even these biogenic elements, are toxic for organisms, if present in high concentrations.

The capability of the aquatic organisms to accumulate heavy metals are tenths and even thousands of times higher than their concentrations in their environment (Gremyachikh et al., 2006, Moiseenko 2003, Podgurskaya et al., 2004). In fact, all water bodies are polluted by heavy metals.

Many of these pollutants possess biological activity and, as opposite to organic compounds, do not undergo transformation in the organisms of aquatic animals. Researches on different types of fish revealed the effect of heavy metals on the physiological activity and biochemical parameters of various tissues of fish (Almeida et al., 2002, Abedi et al., 2013). The ions of Cu, Zn, Hg, and Cd are found to have **morphological and physiological-biochemical parameters in fish**, even at low concentrations (Sarnowski 2003, Sastry et al., 1997, Samson and Shenker 2000, Sastry, and Gupta 1979).

2.4- Pesticides

Extensive application of pesticide in agricultural field affects the enzymatic activity of non-target animals, including fishes. The pesticides affect fish directly by accumulation in their body causes severe impairment in physiologic, metabolic, and structural changes in numerous organs. Fishes can accumulate and retain pesticides from their surroundings. Insecticides are generally accepted as the most acutely toxic class of pesticides, therefore,

pesticide classified as groups organophosphorus compounds (OP), organochlorin compounds (OC), carbamates (CB), pyrethroids, neonicotinoids and botanical pesticides (AnvariFar et al. 2018) .

2.4.1- Effects of pesticide on fish

Pesticides caused reduction of growth and hatching, lower antioxidative response, skeletal deformities, hepatic dysfunction, immunosuppression and finally the death of fish (Cengiz et al., 2017; El-Sayed and Saad, 2008; El-Sayed et al., 2007). The synthetic pyrethroid cypermethrin is used as an insecticide in large-scale agricultural areas. It may be classified as Restricted Use Pesticides (RUP) by the EPA because of cypermethrin's toxicity to fish. The effect of DLM toxicity on hematology, biochemical, and antioxidant status was determined in Nile tilapia (Abdelkhalek et al., 2015).

Other pesticide as fenvalerate was studied recently by Al-Ghanim et al., (2020 who studied the oxidative stress mediated by fenvalerate at sub lethal concentrations after 28 days of exposure of Zebra fish. Following 28 days of exposure of pesticide. Results revealed reduction of superoxide dismutase (SOD) activity after 28 days of exposure in sub lethal concentration of fenvalerate in liver and gills. In liver, catalase activity was found to be less in fenvalerate-exposed fish than control fish. In liver, increase of 75.75% AST and 38% increases in ALT in gills. The findings suggest that in monitoring fenvalerate toxicity programme, enzyme activities can be potent diagnostic tool for fenvalerate induced toxicity (Al-Ghanim et al., 2020) where these enzymes are highly useful marker enzymes to explore the effect of toxic materials such as pesticides to aquatic animals (Chaurasia et al., 2015; Kumaresan et al., 2015b; Rao et al., 2015)

Chapter 3

Industrial pollution

3- Water pollution from Dairy industry

Many of the industries are situated along the banks of river such as dairy products steel and paper industries for their requirement of huge amounts of water in manufacturing processes and finally their wastes containing acids, alkalis, dyes, and other chemicals are dumped and poured down into rivers as effluents. Chemical industries concerning with manufacture of Aluminum release large amount of fluoride through their emissions to air and effluents to water bodies.

Fertilizer industries generate huge amount of ammonia whereas steel plants generate cyanide. Chromium salts are used in industrial process to produce sodium dichromate and other compounds containing chromium. All such discharges finally arrive at water bodies in the form of effluents affecting human health and the organism living there .

3.1 Dairy industry

Globally, the dairy sector is a substantial industry amounting to 900 million tons of milk being produced annually (USDAD, 2022). However, this large-scale production raises concerns about the resulting wastewater. It is estimated that the volume of wastewater generated during milk processing is 2.5 times greater than the volume of milk processed, amounting to a staggering 2.25 billion tonnes of dairy wastewater produced annually (Dogan et al., 2020)

The dairy industry includes the transformation of raw milk into pasteurized and sour milk, yoghurt, hard, soft and cottage cheese, cream and butter products, ice cream, milk and whey powders, lactose, condensed milk, as well as various types of desserts (Abdel-Shafy, & Mansour, 2013) The dairy industries require large quantity of water for the purpose of washing of cans, machinery and floor,

the liquid waste in a dairy originates from manufacturing process, utilities and service section. The clean water is used in various stages of dairy operations, such as, milk processing, cleaning, packaging and cleaning of the milk tankers and releases the wastewater which is known as dairy effluent (Atallah et al., 2020) Water is used for processing in the ratio of 1:10 (water: milk) per liter of milk.

Dairy wastewater has high concentration of dissolved organic components like whey proteins, lactose, fat and minerals (Hussain et al., 2001) and it is also malodorous because of the decomposition of some of the contaminants causing discomfort to the surrounding population. The dairy industry generates huge amount of wastewaters, approximately 0.2–10 L of waste per litre of processed milk (Igneswaran, & Sundaravadivel, 2004).

Dairy wastes are largely neutral or slightly alkaline and have a tendency to become acidic quite rapidly, because of the fermentation of milk sugar to lactic acid. The lower pH may lead to the precipitation of casein. Dairy wastes are characterized by strong butyric acid odor and heavy black flocculated sludge masses (Qadir, et al., 2010). Fats, oil and grease, also called FOG and can have negative impacts on wastewater treatment systems (Okoh et al., 2007). Oil and grease is composed primarily of a fatty matter from animal and vegetable sources, hydrocarbons of petroleum origin, the interferences include sulfur compounds and certain organic dyes (Kolev Slavov 2017).

Organic load is basically constituted by milk (raw material and dairy products), reflecting an effluent with high levels of chemical oxygen demand (COD), biochemical oxygen demand (BOD), oil and grease, nitrogen and phosphorus. Moreover, the automatic cleaning system CIP (cleaning in place)

discards rinse waters with pH varying between 1.0 and 13.0, further complicating the question of treatment (Rosenwinkel et al., 2005).

Wastes from milk product manufacture contain milk solids in a more or less dilute condition, but in varying concentration. These solids enter the waste from almost all of the operations. Wastewater generated by milk-based food industry (Karadag et al., 2015). Contaminants expected in the wastewater from Milk based food industry are shown in table (1).

Dairy wastewater is one of the highly reached organic load wastewater and requires several separation techniques for each milk product to decrease the organic load into accepted limits (Ramadan 2016). The dairy wastewater characteristics and stream are the two most variables that influencing the quality of wastewater medicines.

The dairy industry's wastewater is characterized by a variety of pH values and natural loads spoken to in Chemical Oxygen Request (COD), Organic Oxygen Request (BOD), Add up to Suspended Solids (TSS), Add up to Natural Carbon (TOC) and Lactose (Yu, & Fang, 2002). A few things about were conducted with dairy wastewater treatment utilizing distinctive procedures to control the dairy wastewater into acknowledged limits (Briao, 2000)

	Plant process	Waste generating process	Nature of Waste
1	Milk receiving	Tank truck washing	Milk solids + detergents
2	Clarifying and/ or standardizing	Sludge from centrifugal Milk solids high	Milk solids high in protein & cells
3	Storage of raw milk	Tank washing & sanitizing	Milk solids+ detergents + sanitizer
4	HTST Pasteurization, homogenization	HTST start-up & product change over cleaning	Milk solids + detergent + sanitizer
5	Standardizing, separating, cooling	Sludge from separator	Milk solids high in proteins and cells
6	Vat processing cream	Cleaning and sanitizing	Milk solids+ detergent+ sanitizer
7	Storage of milk + cream	Cleaning and sanitizing	Milk solids+ detergent+ sanitizer
8	Filling and transfer to cooler	Drips, broken package convey or lubrication cleaning	Milk solids + detergents + sanitizer+ lubricant
9	Cold storage & distributor	Broken packages conveyor lubrication returns	Milk solids + detergents + sanitizer + lubricant

Table 3 Sources of pollution in dairy industry (EL-Tawabty et.al. 2022)

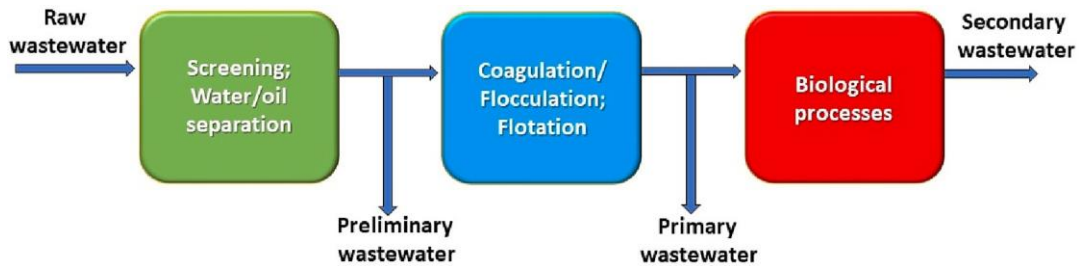


Figure 1 Origin of different types of wastewater from a dairy WWTP.

3.2 Characteristics of dairy industry wastewaters.

Dairy effluent contains soluble organics, suspended solids, trace organics. All these components contribute largely towards their high biological oxygen demand (BODS) and chemical oxygen demand (COD). Dairy wastes are white in colour and usually slightly alkaline in nature and become acidic quite rapidly due to the fermentation of milk sugar to lactic acid.

The suspended matter content of milk waste is considerable mainly due to fine curd found in cheese waste. The pollution effect of dairy waste is attributed to the immediate and high oxygen demand. Decomposition of casein leading to the formation of heavy black sludge's and strong butyric acid odors and characterize milk waste pollution.

The characteristics of a dairy effluent contain Temperature, Color, PH (6.5-8.0), DO, BOD, COD, Dissolved solids, suspended solids, chlorides, sulphate, oil & grease. It depends largely on the quantity of milk processed and type of product manufactured. The waste water of dairy contains large quantities of milk constituents such as casein, inorganic salts, besides detergents and sanitizers used for washing. It has high sodium content from the use of caustic soda for cleaning.

Waste Type	COD	BOD	pH	TSS	TS	References
Milk & Dairy Products factory	10251.2	4840.6	8.34	5802.6		Oneț Cristian,2010
Dairy effluent	1900-2700	1200-1800	7.2-8.8	500-740	900-1350	U. B. Deshannavar, et al 2012
Arab Dairy Factory	3383 ± 1345	1941± 864	7.9 ± 1.2	831 ± 392		A. Tawfik et. al,2007
Dairy waste water	2,500-3,000	1,300-1,600	7.2-7.5	72,000-80,000	8,000-10,000	Javed Iqbal Qazi et. al,
Dairy effluent (CPCB 1993)	1120-3360	320-1750	5.6-8	28-1900		Kusum Lata, et. al, Biogas forum,1999

Whey	71526	20000	4.1	22050	56782	Deshpande D.P. et. al, 2012
Bhandara Co-operative dairy industry wastewater	1400 to 2500	800 to 1000	7.1-8.2	1045 to 1800	1100 to 1600	Monali Gotmare* et al.,2011
Cheese Whey pressed	80,000-90,000	120,000-135,000	6	8000-11000		Rana Kabbout, et al.,2011
Aavin dairy industry washwater	2500-3300		6.4 - 7.1	630-730	1300-1400	Sathyamoorthy G.L, et al.,2012
Dairy industry wastewater	2100	1040	7-8	1200	2500	A. Arumugam

Table 4 Typical Characteristics of dairy industry wastewaters reported by various authors are given in Table

A majority of wastewater gets produced during cleaning operations, especially between products changes when different types of products are produced in a specific production unit and clean-up operations, figure below explains in detail the units involved in milk processing industries and shows the flowchart of effluent sources from various units. (Jai prakash Kushwaha et.al., 2011)

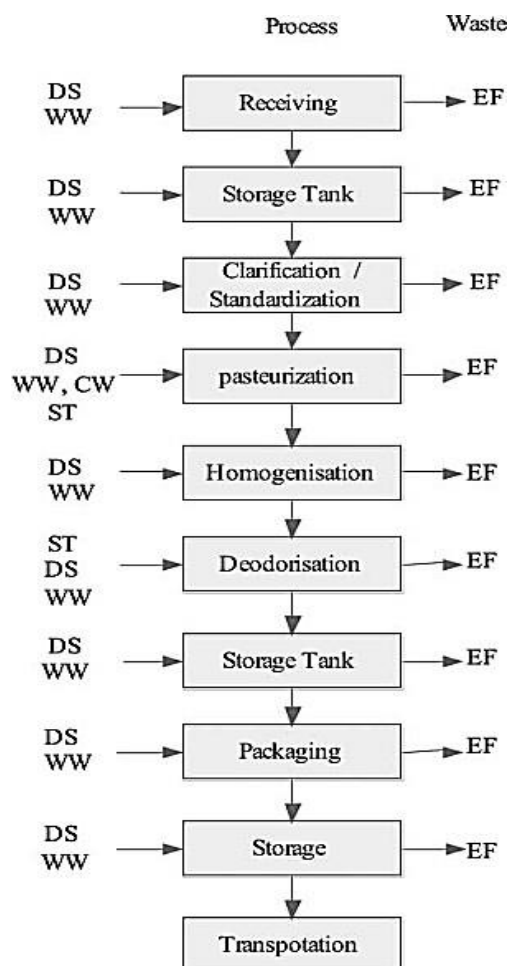


Figure 2 Effluent sources from various units of milk processing. DS-Detergents and Sanitizing Agents, WW- Wash Water, ST-Steam, CW-Cooling Water.

3.3 Effects of Effluents: Environmental Effects.

The dairy industry is one of the most polluting of industries, not only in terms of the volume of effluent generated, but also in terms of its characteristics as well. It generates about 0.2–10 liters of effluent per liter of processed milk with an average generation of about 2.5 liters of wastewater per liter of the milk processed. Dairy

processing effluents are generated in an intermittent way and the flow rates of these effluents change significantly.

The volume, concentration, and composition of the effluents arising in dairy industry are dependent on the type of product being processed, the production program, operating methods, design of the processing plant, the degree of water management being applied, and subsequently the amount of water being conserved. These dairy industries generate different types of waste including: wastewater from the production line (cleaning of equipment and pipes) cooling water, domestic wastewater, the acid whey and sweet.

Due to this the quality and quantity of the product content in the dairy wastewater at a given time changes with the application of another technological cycle in the processing line. The sweet whey form the most polluting effluent by its biochemical composition rich in organic matter (lactose, protein, phosphorus, nitrates, nitrogen) and is from 60 to 80 times more polluting than domestic sewage. The waste water of dairy contains large quantities of milk constituents such as casein, inorganic salts, besides detergents and sanitizers used for washing. All these components contribute largely towards their high biological oxygen demand (BODS) and chemical oxygen demand (COD).

Dairy effluents decompose rapidly and deplete the dissolved oxygen level of the receiving streams immediately resulting in anaerobic conditions and release of strong foul odors due to nuisance conditions. The receiving water becomes breeding place for flies and mosquitoes carrying malaria and other dangerous diseases like dengue fever, yellow fever, chicken guniya.

It is also reported that higher concentration of dairy wastes are toxic to certain varieties of fish and algae. The casein precipitation from waste which decomposes further into a highly odorous black sludge at certain dilutions the dairy waste is found to be toxic to fish also. Dairy effluent contains soluble organics, suspended solids, trace organics.

They decrease DO, promote release of gases, cause taste and odor, impart color or turbidity, promote eutrophication. The main environmental problems related to milk production affect the pollution of water, air and biodiversity. They often cause a growth of algae and bacteria that consume oxygen in the water and eventually suffocate the rivers leading to the gradual disappearance of fish. Hence the need to treat dairy effluents by various processes.

Waste Type	COD	BOD	pH	TSS	TS	References
Milk & Dairy Products factory	10251.2	4840.6	8.34	5802.6		Oneț Cristian,2010
Dairy effluent	1900-2700	1200-1800	7.2-8.8	500-740	900-1350	U. B. Deshannavar, et al 2012
Arab Dairy Factory	3383 ± 1345	1941± 864	7.9 ± 1.2	831 ± 392		A. Tawfik et. al,2007
Dairy waste water	2,500-3,000	1,300-1,600	7.2-7.5	72,000-80,000	8,000-10,000	Javed Iqbal Qazi et. al,
Dairy effluent (CPCB 1993)	1120-3360	320-1750	5.6-8	28-1900		Kusum Lata, et. al, Biogas forum,1999
Whey	71526	20000	4.1	22050	56782	Deshpande D.P. et. al, 2012
Bhandara Co-operative dairy industry wastewater	1400 to 2500	800 to 1000	7.1-8.2	1045 to 1800	1100 to 1600	Monali Gotmare* et al.,2011
Cheese Whey pressed	80,000-90,000	120,000-135,000	6	8000-11000		Rana Kabbout, et al.,2011
Aavin dairy industry washwater	2500-3300		6.4 - 7.1	630-730	1300-1400	Sathyamoorthy G.L, et al.,2012
Dairy industry wastewater	2100	1040	7-8	1200	2500	A. Arumugam

Figure 3 Characteristics of dairy industry wastewaters (composition in mg/l, except pH)

3.4 The impact of dairy industry wastewater on human health.

These Can be extensive and multifaceted, affecting individuals through various pathways and mechanisms. Here's a detailed exploration of these impacts:

3.4.1. Waterborne Diseases:

Dairy wastewater often contains pathogens such as bacteria (e.g., *E. coli*, *Salmonella*), viruses, and parasites (e.g., *Cryptosporidium*, *Giardia*) originating from animal feces, contaminated water used in cleaning processes, and inadequate sanitation practices. When this untreated wastewater enters water sources used for drinking, irrigation, or recreational purposes, it poses significant risks:

3.4.2. Water Contamination:

Pathogens can survive and thrive in untreated wastewater, contaminating groundwater, surface water, and water bodies downstream. This contamination can lead to outbreaks of waterborne diseases among communities relying on these water sources.

3.4.3. Human Exposure:

People can become infected with waterborne pathogens through consumption of contaminated drinking water, ingestion of contaminated foods irrigated with wastewater, or recreational activities in contaminated water bodies.

3.4.4. Health Effects:

Waterborne diseases caused by pathogens in dairy wastewater can manifest as gastrointestinal illnesses (e.g., diarrhea, vomiting), fever, abdominal cramps, and in severe cases, dehydration, and organ damage. Vulnerable populations such as

children, the elderly, pregnant women, and immunocompromised individuals are particularly at risk of severe complications.

3.4.5. Food Safety Concerns:

Crop Irrigation: Dairy wastewater used for irrigation of crops can result in the uptake of pathogens and chemical contaminants by plants. This contamination can lead to food crops that are unsafe for human consumption, potentially causing foodborne illnesses when consumed.

3.4.6. Long-Term Health Risks:

Chronic Exposure: Prolonged exposure to pathogens and chemical contaminants in dairy wastewater can lead to chronic health issues, including recurrent infections, compromised immune function, respiratory problems, and increased susceptibility to other diseases.

3.5 Pathogens in Dairy Wastewater:

Dairy wastewater often contains a variety of pathogens that originate from animal feces, contaminated water used in dairy processing, and inadequate sanitation practices. Common pathogens include: Bacteria: *Escherichia coli* (E. coli), *Salmonella* spp., *Campylobacter* spp., *Listeria monocytogenes*. Viruses: Norovirus, Rotavirus, Hepatitis A virus. Parasites: *Cryptosporidium* spp., *Giardia* spp.

3.6 Transmission and Health Risks:

Water Contamination: Untreated dairy wastewater can contaminate groundwater, surface water, and water bodies downstream. This contamination poses significant risks to human health when people consume contaminated water or use it for daily activities such as bathing and washing.

Food Contamination: If crops are irrigated with untreated dairy wastewater, pathogens can be taken up by plants, leading to foodborne illnesses when these crops are consumed raw or undercooked.

Direct Contact: Individuals can contract infections through direct contact with contaminated water during recreational activities such as swimming or through occupational exposure in dairy farming and processing.

Chemical Contaminants:

Aside from pathogens, dairy wastewater may contain chemical contaminants that originate from:

Antibiotics: Used in dairy farming to treat infections in animals, antibiotics can enter wastewater and contribute to antibiotic resistance in bacteria, potentially impacting human health through reduced effectiveness of antibiotics for treating infections.

Hormones: Growth hormones administered to dairy cattle to increase milk production can also be excreted in waste and enter wastewater, potentially affecting human health through consumption of dairy products or contaminated water.

Cleaning Agents: Dairy processing facilities use detergents, sanitizers, and disinfectants containing chemicals such as chlorine, ammonia, and phosphates. Improper disposal or leakage of these chemicals into wastewater can lead to contamination of water sources and pose health risks to humans.

3.6 Health Impacts:

A. Acute Effects: Waterborne pathogens can cause acute gastrointestinal illnesses such as diarrhea, vomiting, abdominal cramps, and fever. These symptoms can be particularly severe in young children, the elderly, and immunocompromised individuals.

B. Chronic Effects: Chronic exposure to pathogens and chemical contaminants in dairy wastewater can lead to long-term health problems, including recurrent infections, compromised immune function, respiratory issues, and increased susceptibility to other diseases.

The impact of dairy industry wastewater on animals (Toxicity of DWW)

Dairy waste waters are characterized by high biological-oxygen demand (BOD) and chemical oxygen demand (COD) concentrations, and generally contain fats, nutrients, lactose, as well as detergents and sanitizing agents (Javed Iqbal Qazi et al.,2011). Dairy effluents decompose rapidly and deplete the dissolved oxygen level of the receiving streams immediately resulting in anaerobic conditions and release of strong foul odours due to nuisance conditions.

The receiving water becomes breeding place for flies and mosquitoes carrying malaria and other dangerous diseases like dengue fever, yellow fever, chicken guniya. It is also reported that higher concentration of dairy wastes are toxic to certain varieties of fish and algae. The casein precipitation from waste which decomposes further into a highly odorous black sludge at certain dilutions the dairy waste is found to be toxic to fish also. Dairy effluent contains soluble organics, suspended, solids, trace organics (Bharati et al., 2013).

They decrease DO, promote release of gases, cause taste and odour, impart colour or turbidity, promote eutrophication. Due to the high pollution load of dairy wastewater, the milk-processing industries discharging untreated/partially treated wastewater cause serious environmental problems. (Tiwari, 1994). Moreover, the Indian government has imposed very strict rules and regulations for the effluent discharge to protect the environment. Thus, appropriate treatment methods are required so as to meet the effluent discharge standards.

The dairy industry is one of the most polluting of industries, not only in terms of the volume of effluent generated, but also in terms of its characteristics as well (Muthusamy and Jayabalan, 2001). It generates about 0.2–10 liters of effluent per liter of processed milk with an average generation of about 2.5 liters of wastewater per liter of the milk processed (Tiwana, 1985).

Dairy processing effluents are generated in an intermittent way and the flow rates of these effluents change significantly. The volume, concentration, and composition of the effluents arising in dairy industry are dependent on the type of product being processed, the production program, operating methods, design of the processing plant, the degree of water management being applied, and subsequently the amount of water being conserved.

These dairy industries generate different types of waste including wastewater from the production line (cleaning of equipment and pipes) cooling water, domestic wastewater, the acid whey and sweet. Due to this the quality and quantity of the product content in the dairy wastewater at a given time changes with the application of another technological cycle in the processing line. The sweet whey form the most polluting effluent by its biochemical composition rich in organic matter (lactose,

protein, phosphorus, nitrates, nitrogen) and is from 60 to 80 times more polluting than domestic sewage.

The waste water of dairy contains large quantities of milk constituents such as casein, inorganic salts, besides detergents and sanitizers used for washing. Detergents (Sodium hydroxide, Ethylene-diaminetetraacetic acid) represent the biggest portion of chemicals used in dairies; it may be alkaline or acidic. Many Dairy industries have waste treatment facilities such as Effluent Treatment Plant (ETP). Outcome of Effluent Treatment Plant is sludge and scum, a huge problem occurs in disposal of such waste discharged from these industries.

The high values of suspended solids and dissolved solids shows its high pollution potential. Discharge of such wastes into inland surface water will lead to depletion of oxygen in the water bodies, affecting aquatic life and creating unaesthetic anaerobic conditions. All these components contribute largely towards their high biological oxygen demand (BODS) and chemical oxygen demand (COD) (Marwaha et al., 2001). Which is much higher than the specified limits of Indian standard institute (ISI), now Bureau of Indian standard (BIS), for the discharge of

As these wastes are generally released to the nearby stream or land without any prior treatment are reported to cause serious pollution problems. Dairy effluents decompose rapidly and deplete the dissolved oxygen level of the receiving streams immediately resulting in anaerobic conditions and release of strong foul odors due to nuisance conditions impart color or turbidity, promote eutrophication.

The receiving water becomes breeding place for flies and mosquitoes carrying malaria and other dangerous diseases like dengue fever, yellow fever, chicken guniya. It is also reported that higher concentration of dairy wastes are toxic to certain varieties of fish and algae. The casein precipitation from waste which

decomposes further into a highly odorous black sludge at certain dilutions the dairy waste is found to be toxic to fish also

Different categories of toxicity: pH: - It is a term used to express the intensity of the acid or alkaline condition of a solution. It is a way of expressing the hydrogen-ion concentration or the hydrogen-ion activity. Pure water is said to be neutral, with a pH close to 7.0 at 25 °C (77 °F). Solutions with a pH less than 7 are said to be acidic and solutions with a pH greater than 7 are said to be basic or alkaline. The pH of untreated dairy effluent was between 4.5 to 9 and while in industry treated effluent it ranged from 7.0 to 8.5. Though the pH is alkaline in fresh conditions, the waste becomes acidic due to decomposition of lactose into lactic acid under anaerobic conditions and may cause corrosion of sewers (Joseph, 1995).

Colour and Turbidity: Wastewaters that are highly coloured are likely to alter the colour of receiving water. Dairy factory wastes probably contain little soluble colour, although after various forms of treatment true colour may result. Colloidal and particulate components in the waste reflect light back to an observer. This is known as apparent colour. The concept of turbidity is closely related to this phenomenon. Milk wastes contain significant quantities of material that will result in turbidity of discharges (Bharati et al.,2013)

3.7 Water Quality Degradation

Dairy wastewater typically contains high concentrations of organic matter, nutrients (such as nitrogen and phosphorus), suspended solids, and pathogens. When this wastewater is discharged untreated into water bodies, several processes occur:

3.7.1 Oxygen Depletion

The organic matter in the wastewater serves as a food source for bacteria and other microorganisms. As these organisms decompose the organic matter, they consume dissolved oxygen in the water. This process, known as biochemical oxygen demand (BOD), can lead to oxygen depletion in water bodies, especially in stagnant or slow-flowing waters. Aquatic organisms such as fish and invertebrates rely on dissolved oxygen for respiration. Low oxygen levels (hypoxia) can stress or suffocate these organisms, leading to reduced populations or even mass die-offs.

3.7.2. Toxicity

Dairy wastewater may contain various contaminants that are toxic to aquatic life:

A. Heavy Metals: Dairy operations may use metals such as zinc and copper in feed supplements or disinfectants. These metals can accumulate in sediments and bioaccumulate in aquatic organisms, affecting their physiological processes and health .

B. Pharmaceuticals and Hormones: Veterinary pharmaceuticals and hormones used in dairy farming can enter wastewater through excretion or improper disposal. These compounds can disrupt endocrine systems in aquatic organisms and affect their reproduction and development.

3.7.3. Bioaccumulation

Some contaminants in dairy wastewater have the potential to bioaccumulate in aquatic organisms. This means that even low concentrations of persistent pollutants, such as certain pesticides or industrial chemicals, can accumulate in the tissues of organisms over time. Predators higher up the food chain, including birds and mammals that consume contaminated fish or invertebrates, may experience biomagnification, where concentrations of contaminants increase with each trophic level, posing health risks to these animals .

3.7.4. Eutrophication

The nutrient-rich nature of dairy wastewater can contribute to eutrophication of water bodies. Excess nitrogen and phosphorus promote algal blooms, particularly of cyanobacteria (blue-green algae). These blooms can produce toxins harmful to aquatic organisms and humans (Paerl and Otten, 2013). Moreover, when the algae die and decompose, the further process depletes dissolved oxygen, exacerbating hypoxic conditions.

3.7.5. Impact on Terrestrial Animals

Dairy wastewater, if not properly managed, can contaminate soil and groundwater. Livestock and wildlife may ingest contaminated water or crops irrigated with it, leading to health issues such as antibiotic resistance or direct toxicity from chemical contaminants.

Chapter 4

Evaluation the dairy wastewater on tilapia (*O. niloticus*)

4- Evaluation the dairy wastewater on tilapia (*O. niloticus*)

4.1 Material and methods :

- Source of the dairy wastes.

The dairy wastes were obtained from EL Tayeb CO. for Dairy industry; Industrial zone Gamasa, El Dakahleya , Egypt.

- The chemical composition of DWW

the chemical composition of the dairy wastewater Sample was analyzed in The central laboratory at the Faculty of Agriculture, Mansoura University, which is internationally accredited. Dry mater, protein, fat and ash was determined

Fish and culturing conditions

This experiment is designed to evaluate the effects of dairy wastes on survival rate, stress markers of Nile tilapia (*O. niloticus*) during 96 h of exposure. Sixty fingerlings of *O. niloticus*, with a mean weight of 40 ± 0.2 g, were sourced from a private fish farm in Al-Manzala, Dakahlia Province. Tilapia fish were randomly divided into 4 groups (T_0 – T_3) and were acclimated to the aquarium conditions for 2 weeks before the experiment was initiated. Each group contained fifteen fish which were maintained in four glass aquaria (70 x 40 x 30 cm) in closed system (recirculation system RAS), according to the design of experiment (Table 1) Serial dilutions of dairy wastes were prepared in dechlorinated tap water. Crude dairy wastes were used in different dilutions 0, 5, 10, and 25 %) each glass fish aquaria containing 40 L (of the aerated different tested dilutions). Behavior pattern and mortality percentage of fish groups were noted over 96 hr. The water temperature was about 25 ± 2 °C, the aquaria were provided with air stones,

and all fish were fed twice daily at a feeding rate of 3 % of the actual body weight. The experiment was designed as follows:

Groups	% Dairy wastewater	Dilution rate% by water tap
Control group (T₀)	0	100
T1	5	95
T2	10	90
T3	25	75

Table 5 Table: Experimental fish groups

Sample collection

At the end of the experiment, blood samples were withdrawn from the fish heart of each group to investigate stress biochemical indices. After centrifugation at 3500 ×g for 15 minutes at 4 °C, the serum were separated, recovered, and stored at -20 °C for analysis

Stress markers assessments

Cortisol concentration in serum, were evaluated utilizing test kits supplied by Cusabio Co. (Texas, USA), Moreover, The glucose level was determined using altered methods (Wedemeyer and Yasutake, 1977).

The survival rate (SR%) was determined as the number of fish at the end of the experiment/initial number of fish ×100 (Lugert et al., 2016).

4.2 Results and Discussion

	Dry mater %	Total protein	Nitrogen	Ash	Moisture
DWW	20.83	5.5	0.86	63.38	79.71

Table 6 The chemical composition of the DWW %is presented in Table

Table (6) showed that the Dairy waste waters are characterized by high value of ash, salinity (63.38%) which led to stress on the fish

Assessing the harmful dose of DWW on survival rate %

Mortality Rate: Adverse effects the dairy wastes on the survival rate was reported after 48 and 96 hr. (Table 2). The control group T0 had 100% survival and displayed no unusual behaviours when exposed to 0 DWW (table 3). Fish in T1 had the lowest mortality rate (13 & 26 % at 48 and 96h respectively), while the mortality rate gradually increased by increasing the percentage of the dairy wastes in the fish groups T2 and T3 with the least amount of resting activity, the highest surfacing frequency, surface swimming, laterality, and aggression behavior).

	% of DW	At 48 hour	96h
Control group (T₀)	0	0	0
T1	5	13	26
T2	10	20	33
T3	25	40	73

Table 7 : Mortality rate (%) of fish exposed to different dilutions of dairy wastes

Stress-related parameters

Table 5 shows that there were discernible variations in the cortisol and glucose levels between groups. When comparing the DWW groups (T1-T3) to the T0 group, the T3 had remarkably higher levels of cortisol and glucose, followed by the T2 group.

	% of DW	Cortisol	Glucose
Control group (T₀)	0	8.96	29
T1	5	13.5	52
T2	10	19.7	43
T3	25	22.6	40

Table 8 variations in the cortisol and glucose levels

Cortisol and glucose levels are frequently valid markers for the investigation of fish's physiological response to many environmental stressors through regulating the organism's growth, metabolic energy, hemostasis, and immune capability (Zahran et al., 2021a, b), subsequently, elevated cortisol levels may have adverse effects on these previously mentioned responses (Barton, 2002). To lessen the negative impacts of stress on aquatic species, blood glucose levels are a secondary response that is often sequestered for further energy generation (Dossou et al., 2018). . Due to those measurements are generally accurate indicators of different types of stress, it was revealed that heat exposure caused metabolic stress in Rohu, *Labeo rohita* fish (Ciji and Akhtar, 2021; Ciji et al., 2012). Additionally, cortisol and glucose levels of African catfish cultured in high heat stress conditions were significantly elevated (Dawood et al., 2022). As well, higher glucose and cortisol levels were seen in striped catfish (*Pangasianodon hypophthalmus*) cultivated in hypersalinity and high temperatures (Phuc et al., 2017). The increase in glucose

levels might be associated with stress, as stress has been demonstrated to trigger the release of catecholamines, leading to the accelerated breakdown of glycogen and the subsequent rise in blood glucose levels (Nakano and Tomlinson, 1967).

Control of dairy wastewater (DWW).

The release of DWW into water systems without proper treatment causes an array of environmental issues due to its high organic content. The decomposition of DWW in a water source causes the dissolved oxygen (O₂) inside the water body to drop as the microbes (especially algae) flourish in the nutrient-rich environment.

This results in a hypoxic environment which is unsuitable for aquatic animal and plant life. The increased algal growth produces elevated levels of CO₂ which lowers the pH of the body of water, which in turn slows the growth of fish and shellfish. The addition of DWW into fresh water also provides a breeding ground for various insects, which can upset the balance of the surrounding ecosystem. The degradation of casein also causes the formation of a sludge which is harmful to aquatic organisms. This impacts the health of aquatic life and the potential for fishing which has a negative economic impact. In addition to these environmental implications, the processing of dairy products and the release of DWW can lead to several social impacts. These stem from two main sources, namely, the discharge of untreated or partially treated wastewater into nearby water bodies and the odour produced by the plant and effluents. The contamination caused by the release of DWW can promote the spread of bacteria such as *Salmonella* and *E. coli*, which can cause illness in humans and animals which use

the water sources. Once contracted, these illnesses can easily be spread throughout the community.

Conclusions.

The main sources of pollution in natural water are agricultural runoffs, industrial effluents, and municipal wastewater which are dumped recklessly into natural water and gradually render their water unfit for aquatic life (Abdel Satar et al. 2017).. Human manufacturing operations are the cause of a wide variety of hazardous substances. Once they penetrate natural environments, they change the geochemistry of large areas and the consistency of the physicochemical parameters of water bodies (Reid et al., 2018). Anthropogenic contamination in aquatic environments impacts all elements of the food chain: it changes the ecological composition of phytoplankton and the distribution and abundance of macro-invertebrates (Collins and Fahrig, 2020). Radioactive waste is another type of pollution. It's produced from nuclear power plants, uranium mining, and military weapons development and testing, as well as from universities and hospitals using radioactive materials for medicine and research. For thousands of years, nuclear waste will remain in the environment making disposal a major challenge. Accidentally released pollutants threaten marine resources, groundwater and surface water (Annamalai and Arunachalam, 2017). The various pollutants including pharmaceutical compounds eliminated directly or indirectly into the natural water affect the water quality of water body. The pharmaceutical compounds and their metabolites are released in wastewater treatment plants through urine and feces (Soliman and Sayed 2020). There is great confusion about the exact role of pollution on fish health. There is clear evidence, however, that long-term exposure to certain pollutants is having a detrimental effect on the health of most fish species. As a result of natural accidents, as a breakdown of algal blooms or/and as a result of human activities, pollutants can enter the water environment, resulting in a harmful effect on water quality (Zhelev et al., 2020).

5- Conclusion

High levels of certain contaminants can be directly responsible for the death of large numbers of marine species, such as oil spills. The presence of organic materials in water, such as faecal matter, can therefore increase the amount of microbes that can cause fish diseases. Some contaminants may cause stress on aquatic organisms that may lead to increased pathogen exposure. Fish are exposed to pesticides and their residues and heavy metals by various routes such as runoffs or spray drifting from agricultural fields (Maurya et al., 2019).

Fish serve as a bio-indicator of the aquatic ecosystem to assay the toxicity of pollutants. Ecotoxicology includes various levels of biological organization, including the immune system that is almost common to all multicellular organisms and serves as a direct connection between an organism and its environment. Its impairment will cause fish quality deterioration and disease susceptibility. The abiotic factors such as the metals concentration and pesticides levels in the water suppress humoral and cellular components of the immune system of fish (Banday et al., 2020)

Aquatic pollution is a main problem in many marine and freshwater in worldwide. Stress conditions are responsible for fish disease in aquaculture, which leads to disease. In addition, Water pollution increases sensitivity to infectious diseases, leading to higher mortality rates. (Sarmiento et al., 2004) because any variation in water quality leads to conditions that allow diseases to occur (Irgeui et al., 2015) water pollution has become a major concern for aquatic organisms such as fish due to the increasing the activity of the industrial and agriculture sectors.

Heavy metal pollution is one of the main important sources of water pollutants (Singer et al. 2005) that are continuously in the environment and have a capability to accumulate by fish (Veena et al. 1997). Many studies have focused on the impacts of pollutants on the physiological and immunological function of fish (Halpern et al., 2008), documenting changes that range from slight alteration in morphology and physiology (Guardiola et al., 2015) to increases in sensitivity to disease and even mortality (Al-Ansi et al., 2002). **The aim of this investigate to study the impacts of the Impact of dairy wastewater on the Nile tilapia in vitro.**

Recommendations

Recommendations

1- Literature Review.

- Investigate existing studies on the effects of wastewater on aquatic organisms, specifically focusing on dairy product wastewater.
- Explore research on the physiology and ecology of *Tilapia nilotica* to understand their sensitivity to pollutants.

2- Wastewater Characterization.

- Conduct a detailed analysis of the dairy wastewater, including chemical (e.g., pH, COD, BOD, nutrient levels), physical (e.g., temperature, turbidity), and biological (e.g., microbial load) parameters.

3- Experimental Setup.

- Design controlled experiments with different concentrations of dairy wastewater to observe the effects on *Tilapia nilotica*.
- Use control groups with clean water to compare the results.

4- Health and Growth Monitoring.

- Monitor the health of the tilapia by measuring growth rates, weight, and size.
- Assess any physiological changes, such as gill and liver histopathology.

5- Water Quality Parameters.

- Regularly measure water quality parameters in the experimental tanks, including dissolved oxygen, ammonia, nitrites, nitrates, and phosphorus levels.

6- Behavioral Studies.

- Observe any behavioral changes in tilapia, such as feeding habits, swimming patterns, and stress responses.

7- Reproductive Effects.

- Investigate the impact on reproductive parameters, including spawning frequency, egg quality, and fry survival rates.

8- Microbiological Analysis.

- Study the microbial communities in the wastewater and their potential effects on fish health.

9- Toxicological Assessments.

- Conduct toxicological assessments to determine the lethal and sub-lethal concentrations of dairy wastewater for **Tilapia nilotica**.

10- Field Studies.

- Complement laboratory experiments with field studies to assess real-world impacts in natural or semi-natural environments.

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